# SOL Data Cleaning Studio — Processing & Audit Report

Filename: customer_data_2026.csv | Date: 2026-06-20 04:39:24 | Task ID: test-task-123

## 1. Technology Stack & Execution Context

The complete execution configurations, library list, and orchestration parameters are documented below:

| | |
|---|---|
| Execution Platform / Environment | Local Fallback Engine |
| Remote Kaggle Kernel URL | N/A |
| Kaggle Orchestrator Account | al_dalil_governance_service |
| Cleaning Strategy Inferred | Gamma |
| User's Custom Cleaning Goal | Clean the missing values, format email addresses, and detect salary outliers. |
| Core Engineering Libraries | pandas, numpy, reportlab, python-bidi, arabic-reshaper, scikit-learn |
| Processing Execution Status | COMPLETED |

## 2. Executive Cleaning Statistics

Comparative analysis of dataset shape, counts, and null cells before and after processing:

| | |
|---|---|
| Initial Rows Count | 15,230 |
| Cleaned Rows Count | 14,980 |
| Initial Missing Cells | 2,340 |
| Remaining Missing Cells | 12 |
| Total Rectified Cells | 2,328 |
| Truth Confidence / Quality Score | 98.4% |

## 3. Enterprise Quality Guardrails

The automated validation guardrails check for data cleaning execution yields the following status:

| | |
|---|---|
| Blocked Actions / High-Risk Operations | • Column 'SSN': modification blocked due to privacy policy |
| Guardrail Warnings | • Detected high null value percentage in column 'Age' |
| Permitted & Completed Actions | • Imputed missing numerical values in column 'Age' using median<br>• Converted column 'Email' values to standardized lowercase<br>• Removed 250 rows containing duplicate primary keys |

## 4. Detailed Anomalies Audit Log

The table below lists every single anomaly detected in the dataset and how it was resolved:

| Target Column | Issue / Anomaly Detected | Resolution / Imputation Method |
| --- | --- | --- |
| Age | Missing values | AI Imputation (Median) |
| Email | Non-standard uppercase letters | Standardized to lowercase |
| Salary | Outlier values detected (Z-Score > 3) | Capped outlier values to 99th percentile |
| DateOfBirth | Inconsistent date formats | Parsed and formatted to ISO-8601 YYYY-MM-DD |